

A Bounded Causal Entropy Tax

Unequal block lengths remove the growing boundary cost

Mathematical construction for the C3-C model

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Abstract

For any two finite output laws with strictly different Shannon entropies, we construct exact finite-horizon causal realizations whose seed entropy is $n \max\{H(P), H(Q)\} + O(1)$. The constant is independent of the horizon and the construction is valid for every adaptive action policy. An explicit example has a strictly positive residual bounded by 14 bits for every n . The construction uses geometrically increasing and decreasing block lengths in the anti-diagonal arrangement. This result concerns a strict entropy gap; it does not resolve the equal-entropy logarithmic problem.

1 Statement and model

All logarithms and entropies are base two. A discrete seed fixes a history-indexed deterministic response tree. At each step an action selects one of two finite laws P, Q . Under every adaptive policy the outputs must have exactly the prescribed conditional laws. Let $C_n^{\text{causal}}(P, Q)$ be the infimum of the seed entropy. This is the C3-C model of the supplied *Who Pays the Bill?*, Version 4; no synchronous cross-history identity is assumed. Zero-probability symbols may be discarded.

Theorem 1 (Bounded tax under a strict entropy gap). *If $H(P) > H(Q)$, there is a finite constant $K(P, Q)$ such that, for all n ,*

$$\boxed{nH(P) \leq C_n^{\text{causal}}(P, Q) \leq nH(P) + K(P, Q).}$$

Moreover, $C_n^{\text{causal}} - nH(P)$ is nondecreasing and has a finite limit. The same assertion holds after exchanging P, Q .

Theorem 2 (An explicit nonzero bounded residual). *Let P be uniform on eight symbols and let $Q = (1/3, 2/3)$. For every positive integer n ,*

$$\boxed{\frac{h_2(1/3)}{8} \leq C_n^{\text{causal}}(P, Q) - 3n < 14,} \quad \frac{h_2(1/3)}{8} = 0.1147869792568\dots,$$

where h_2 is binary entropy. Thus the residual has order $\Theta(1)$, with a strictly positive finite limiting value.

The exact value of that limit is not determined here. The upper constant 14 is deliberately loose. Different action alphabets cause no difficulty; one may pad Q with zeros if a common alphabet is required.

2 The construction: variable-length anti-diagonal blocks

Choose positive integers $s_1, \dots, s_m$ with sum n . Partition both length- n streams X, W into consecutive blocks of these lengths. Draw the following components independently:

$$X^{(1)} \sim P^{\otimes s_1}, \quad W^{(m)} \sim Q^{\otimes s_m}, \quad (W^{(j)}, X^{(j+1)}) \sim J_j \quad (1 \leq j < m),$$

where J_j is any coupling of $Q^{\otimes s_j}$ and $P^{\otimes s_{j+1}}$. On the a th occurrence of action P , output X_a ; on the b th occurrence of action Q , output W_{n+1-b} .

Lemma 3 (Causal validity and entropy accounting). *This construction is exact under every adaptive policy. If $H(J_j) \leq s_{j+1}H(P) + d_j$, then its seed satisfies*

$$H(S) \leq nH(P) + s_mH(Q) + \sum_{j=1}^{m-1} d_j. \quad (1)$$

Proof. Put $T_j = s_1 + \dots + s_j$. Seeing any coordinate of $X^{(j+1)}$ requires at least $T_j + 1$ occurrences of P . Seeing any coordinate of $W^{(j)}$ in reverse order requires at least $n - T_j + 1$ occurrences of Q . Seeing both therefore requires $n + 2$ actions, which is impossible. For a fixed action word, each independent seed component contributes coordinates from at most one of its marginal product blocks. The observed outputs consequently have the required product law. An output branch of an adaptive deterministic policy fixes one such action word; applying the same statement to that branch proves exactness. Randomized policies follow by conditioning on their independent randomness.

Independence of the seed components gives $H(S) = s_1H(P) + s_mH(Q) + \sum_j H(J_j)$, proving (1). $\square$

Unlike equal-length blocking, this formula only pays for the *last* Q -block as an uncovered boundary. Both boundary blocks can remain small while the interior block lengths become large. A strict entropy gap permits the necessary decrease of block lengths on the far side of the construction.

3 A complete elementary proof of the explicit example

Lemma 4 (Uniform fine grid). *If A is uniform on M symbols and B has $K \geq 2$ symbols, there is an exact coupling with*

$$H(A, B) \leq \log_2 M + \frac{K-1}{M} \log_2 K.$$

Proof. Use the common-uniform interval coupling on $[0, 1)$. The A -intervals all have length $1/M$. Only cells containing an internal endpoint of a B -interval can have positive conditional entropy $H(B \mid A = a)$. There are at most $K - 1$ such cells, and each contributes at most $(1/M) \log_2 K$. Endpoints on grid boundaries do not split cells. $\square$

Apply the lemma to $A = P^{\otimes p}$, $B = Q^{\otimes q}$. If $p \geq q/2$,

$$H(A, B) - 3p \leq q 2^{q-3p} \leq f(q), \quad f(q) = q 2^{-q/2}. \quad (2)$$

No approximation of either marginal is involved.

An integer mountain for every horizon. For $n \geq 2$, choose the unique $k \geq 0$ such that $4 \cdot 2^k \leq n + 2 < 8 \cdot 2^k$, and put $s = 2^k$. Use the left tail $(1, 2, \dots, 2^{k-1})$ and its reverse as the right tail; for $k = 0$ the tails are empty. The remaining length is

$$R = n - 2(s - 1), \quad 2s \leq R < 6s.$$

Set $t = \lceil R/s \rceil$, and divide R as evenly as possible into t integer central blocks. There are at most six central blocks. Each has length between $s/2$ and s (and length 1 if $s = 1$). The resulting sequence sums to n , begins and ends at 1, and satisfies

$$s_{j+1} \geq s_j/2 \quad \text{for every adjacent pair.}$$

Thus (2) applies to every coupling in Lemma 3. Each dyadic tail length appears at most twice, so

$$C_n^{\text{causal}} - 3n \leq H(Q) + 2 \sum_{j=0}^{\infty} f(2^j) + 6 \max_{q \in \mathbb{N}} f(q) < 14. \quad (3)$$

Here are elementary certificates for the last inequality. The sequence $f(q)$ is maximized at $q = 3$, since $f(q+1)/f(q) = (1 + 1/q)/\sqrt{2}$. Hence

$$\max_q f(q) = \frac{3}{2\sqrt{2}} < \frac{16}{15}.$$

The first five terms of the dyadic series are $1/\sqrt{2}, 1, 1, 1/2, 1/16$. The remaining tail is smaller than $1/1024$ (its first term is $1/2048$, with subsequent ratios below $1/2$). Therefore

$$\sum_{j \geq 0} f(2^j) < \frac{71}{100} + 1 + 1 + \frac{1}{2} + \frac{1}{16} + \frac{1}{1024} < \frac{33}{10}.$$

Since $H(Q) < 1$, the right side of (3) is strictly below $1 + 2(33/10) + 6(16/15) = 14$. The unrounded expression in (3) is approximately 13.82244702.

Strict positivity, not a relabeling trick. At horizon one, a seed can be replaced by its pair of action outputs without increasing entropy. Write x_i for the joint probability of the i th P -symbol and the first Q -symbol. Then

$$0 \leq x_i \leq 1/8, \quad \sum_{i=1}^8 x_i = 1/3, \quad H(P, Q) = 3 + \frac{1}{8} \sum_{i=1}^8 h_2(8x_i).$$

A concave function on this polytope has a minimizer at an extreme point. There, at most one x_i is strictly between 0 and $1/8$. Since $8/3 = 2 + 2/3$, the unique fractional row has conditional probability $2/3$. Consequently

$$C_1^{\text{causal}} = 3 + \frac{1}{8} h_2(1/3).$$

For any realization of horizon $n + 1$, first choose action P . Conditional on that first output, the remaining seed is an admissible horizon- n seed. The chain rule gives

$$C_{n+1}^{\text{causal}} \geq H(P) + C_n^{\text{causal}}. \quad (4)$$

This proves the lower bound in Theorem 2, and also handles $n = 1$ in the upper bound. A positive, nondecreasing, bounded residual has a positive finite limit. This completes the explicit example.

4 Why every strict entropy gap suffices

Write $h_P = H(P) > h_Q = H(Q) > 0$. Choose a constant $1 < \Lambda < h_P/h_Q$. The case $h_Q = 0$ is simpler: Q is deterministic, and a length- n iid P stream gives exactly $C_n^{\text{causal}} = nh_P$.

Lemma 5 (Exponentially small block excess). *There are constants $C, c > 0$, depending only on P, Q, Λ , such that for all positive integers p, q with $1/\Lambda \leq p/q \leq \Lambda$, an exact coupling satisfies*

$$H(P^{\otimes p}, Q^{\otimes q}) \leq ph_P + F(q), \quad F(q) = C(q+1)e^{-cq}.$$

Proof. Choose $\varepsilon > 0$ with $(h_P - \varepsilon)/\Lambda > h_Q + \varepsilon$. Except on an event exponentially small in p , a P -word has probability at most $u = 2^{-p(h_P - \varepsilon)}$. Except on an event exponentially small in q , a Q -word belongs to a set of at most $K = 2^{q(h_Q + \varepsilon)}$ words. These exponential estimates follow directly from the exponential Markov inequality for sums of bounded independent information contents: the centered log moment-generating function has derivative zero at the origin, so any fixed nonzero deviation has a strictly positive exponential rate. If an information content is constant, its exceptional event is empty.

Arrange the typical Q -words consecutively in the common-uniform interval coupling. Decode a Q -word from a P -word whenever the latter's interval is contained in a typical Q -interval; otherwise choose an arbitrary decoder output. Decoder error can occur only on an atypical P -word, an atypical Q -word, or a typical P -interval crossing one of the at most $K+1$ typical-region boundaries. Thus its probability is at most

$$\delta \leq e^{-c_1 p} + e^{-c_2 q} + (K+1)u \leq C_1 e^{-c_3 q},$$

with constants adjusted if an exceptional event is empty. For the actual error probability $e \leq \delta$, the elementary decoding entropy bound gives

$$H(Q^{\otimes q} \mid P^{\otimes p}) \leq h_2(e) + e q \log_2 |\text{supp } Q|.$$

This is bounded by $C(q+1)e^{-cq}$ for all sufficiently large q . Increase C to cover the finitely many smaller integer pairs. The coupling itself has exact marginals throughout; typical sets are used only to estimate its entropy. $\square$

A mountain with arbitrarily gentle slopes. Choose $1 < \lambda < \Lambda$. Pick positive integers M, b so large that

$$\lambda + 1/b \leq \Lambda, \quad \frac{M}{M+1} - \frac{1}{b} \geq \frac{1}{\Lambda},$$

and put $a_j = \lceil b\lambda^j \rceil$. Then $a_{j+1}/a_j \leq \Lambda$ and $a_j \geq b\lambda^j$. For $n \geq Mb$, choose the largest integer $k \geq 0$ with

$$n \geq 2 \sum_{j < k} a_j + Ma_k.$$

Use the two tails $(a_0, \dots, a_{k-1})$ and its reverse. For the remaining length R , maximality gives

$$Ma_k \leq R < 2a_k + Ma_{k+1} \leq (2 + M\Lambda)a_k.$$

Divide R evenly into $t = \lceil R/a_k \rceil$ integer central blocks. Each central length ℓ satisfies

$$\frac{a_k}{\Lambda} \leq a_k \frac{M}{M+1} - 1 \leq \ell \leq a_k.$$

There are at most $T = \lceil 2 + M\Lambda \rceil$ such blocks. All successive ratios of the entire sequence lie in $[1/\Lambda, \Lambda]$. Its last block is b if $k \geq 1$, and at most b if $k = 0$.

Apply Lemmas 3 and 5. Uniformly in $n \geq Mb$,

$$C_n^{\text{causal}} - nh_P \leq bh_Q + 2 \sum_{j=0}^{\infty} F(a_j) + T \sup_{q \geq 1} F(q) < \infty. \quad (5)$$

The series converges because a_j grows geometrically and F decreases exponentially at infinity. For the finitely many horizons $n < Mb$, use independent iid streams and enlarge the constant by at most Mbh_Q . The constant-action P path supplies the lower bound nh_P . Equation (4) gives monotonicity, completing the proof of Theorem 1.

5 What this does and does not settle

The strict entropy gap supplies enough excess information inside a long P -block to absorb a somewhat longer Q -block at exponentially small additional entropy cost. Growing and then shrinking the blocks leaves bounded endpoint lengths and a summable sequence of coupling costs. This is the step that equal-size blocks miss.

When $H(P) = H(Q)$, the interval $(1, H(P)/H(Q))$ is empty and the argument cannot be used. In particular, it does not override the square-root lower bound for the equal-entropy, unequal-entropy crash-test pair, nor prove a logarithmic bound for a variance-matched equal-entropy pair. The constant in (5) need not remain bounded as the entropy gap tends to zero. The constructed seed may depend on n ; no single projectively consistent infinite seed is asserted.

The anti-diagonal mechanism is inherited from Section 18 of the user-supplied manuscript by J. Mikulík, *Who Pays the Bill?*, Version 4 (19 August 2026). This note supplies the variable-length construction and its entropy estimates with complete proofs. No claim of historical priority is made. The accompanying Python script verifies integer mountain partitions, exact rational block marginals, and the rational certificates for the explicit bound; the all-horizon and adaptive claims are proved above rather than inferred from those tests.